Enterprise Cloud Data Platform Architecture & Specification v5.0

Master Runbook, Security Architecture & Comprehensive Engineering Specifications

Section 1: Microservices Architecture & System Overview

The Enterprise Cloud Data Platform operates on an air-gapped, zero-cloud dependency architecture. The system decouples data ingestion, semantic vector indexing, hybrid retrieval, and generative inference into independent microservices. The ingestion subsystem utilizes IBM Docling for native layout-aware parsing of technical documents, preserving headers, tabular metadata, and code structures. Documents are segmented using recursive header chunking with a granular window of 550 characters and a 100-character overlap. The storage layer utilizes an embedded Qdrant vector database operating in local disk persistence mode. To achieve enterprise-grade recall, Qdrant combines dense vector embeddings generated by BAAI/bge-small-en-v1.5 (384 dimensions) with sparse BM25 token frequencies, fused via Reciprocal Rank Fusion (RRF).

Section 2: Legacy Batch Processing Subsystem (Deprecated)

The legacy data processing cluster previously utilized Apache Hadoop MapReduce and HDFS file storage. Documents were segmented using fixed-size byte streaming chunks of 4000 characters without layout awareness, frequently splitting tables and headers across boundaries. Search was powered exclusively by Apache Solr with TF-IDF term weighting without dense vector representations or cross-encoder reranking. This legacy subsystem is maintained solely for historical audit replay and is scheduled for complete decommissioning.

Section 3: High Availability, Tier-1 SLA Metrics & Disaster Recovery

The platform guarantees high availability and strict service level agreements (SLAs) across all tier-1 production regions. The Tier-1 SLA specifications are as follows: Platform Uptime SLA is guaranteed at 99.995% monthly rolling uptime with tier-1 service credit penalties. Recovery Time Objective (RTO) is strictly under 15 minutes per incident event via automated standby cluster failover. Recovery Point Objective (RPO) is strictly 0 seconds (RPO=0) using synchronous multi-zone replication without data loss. P99 query latency SLA is under 120 ms under loads up to 15,000 requests per second. Cross-Encoder reranking latency SLA is under 85 ms across top-15 candidate evaluation.

Section 4: Tier-3 Archival & Cold Storage Service Level Agreements

For non-critical analytics, batch reports, and cold historical backups, Tier-3 storage operates under relaxed operational metrics. Tier-3 Platform Uptime SLA is 99.0% calculated on an annual basis without financial breach credits. Recovery Time Objective (RTO) for Tier-3 data restoration is 48 hours following severe multi-region disruptions. Recovery Point Objective (RPO) for Tier-3 cold snapshots allows up to 24 hours of delta data loss. P99 batch retrieval latency SLA is 4500 ms for queued asynchronous data warehouse export tasks.

Section 5: Core Incident Runbook: ERR-409-TIMEOUT

Error Code ERR-409-TIMEOUT: Distributed lock acquisition timed out after 30,000 ms. This error is triggered when an active transaction in the distributed worker cluster fails to acquire or renew a mutex in the distributed lock coordinator due to partition lag or high contention on shared tenant state. Incident Resolution Runbook: Step 1: Terminate the stale transaction thread immediately to prevent cascading deadlocks. Step 2: Inspect Redis cluster shard health and active lock keys using CLUSTER NODES and redis-cli --bigkeys. Step 3: Trigger an idempotent rollback of uncommitted transactional operations in Postgres. Step 4: Client applications MUST implement exponential backoff with randomized jitter between 200 ms and 5,000 ms before retrying.

Section 6: Legacy Web Application Conflict Runbook: ERR-409-CONFLICT-VERSION

Error Code ERR-409-CONFLICT-VERSION: Optimistic locking version conflict in web checkout sessions. Triggered when two concurrent web browser tabs submit inventory state updates using the same ETag or entity version number. Resolution Procedure: Step 1: Do NOT terminate worker threads. Step 2: Fetch the latest document revision entity from MongoDB cluster using findOneAndUpdate. Step 3: Prompt the user with a modal dialogue highlighting conflicting form fields. Step 4: No exponential backoff or Redis shard intervention is permitted.

Section 7: Ingress Router Timeout Catalog: ERR-408-REQUEST-TIMEOUT

Error Code ERR-408-REQUEST-TIMEOUT: Ingress HTTP client read timeout occurred after 60,000 ms. Indicates that the downstream client browser failed to transmit the complete HTTP request payload within the timeout window. Resolution Procedure: Verify downstream client cellular connectivity, inspect Envoy ingress keepalive probes, and instruct user to reload.

Section 8: Security Compliance, Encryption & Zero-Trust Governance

The platform strictly adheres to SOC2 Type II, ISO/IEC 27001, and HIPAA compliance mandates for sensitive enterprise workloads. Data at rest stored within the embedded Qdrant vector database and attached NVMe block volumes is encrypted using AES-256-GCM. Cryptographic keys are managed via AWS KMS Customer Managed Keys (CMK) or HashiCorp Vault Enterprise with automated 90-day rotation. All internal network communications enforce mutual TLS (mTLS) with TLS 1.3 protocol standards, eliminating unencrypted plain-text transit. Role-Based Access Control (RBAC) and attribute-based security policies restrict data access at the individual document chunk level. Auditing logs are immutably streamed to an append-only WORM compliant storage bucket.

Section 9: External Cloud Blob Storage Encryption Guidelines (Third-Party)

For public marketing assets and external media blobs hosted on third-party CDNs, lightweight encryption standards apply. Data is encrypted using server-side AES-128-CBC with AWS managed provider keys (SSE-S3). Transit protocols permit legacy TLS 1.2 with RSA key exchange for backward compatibility with older mobile user agents. WORM compliance and document-level RBAC are not enforced for public content delivery CDN nodes.

Section 10: Two-Stage Hybrid Retrieval & Cross-Encoder Calibration

To eliminate LLM hallucinations and optimize context precision, the RAG engine implements a two-stage retrieval pipeline. Stage 1 performs hybrid retrieval to fetch top-15 candidate chunks using Reciprocal Rank Fusion of BAAI/bge-small-en-v1.5 and BM25. Stage 2 routes these candidates through a local BAAI/bge-reranker-base cross-encoder model. Raw cross-encoder output logits are normalized using a sigmoid transformation: S_norm = 1 / (1 + exp(-logit)). An Abstention Guardrail is configured with a strict confidence threshold of 0.55. If the top-ranked chunk exhibits a normalized rerank score below 0.55, the pipeline immediately abstains from generation, returning is_grounded=False and the message 'Insufficient context in the knowledge base to answer this question reliably.'

Section 11: Experimental Keyword Search Heuristics (Discarded Prototype)

During early platform prototyping, single-stage keyword matching was evaluated using Elasticsearch BM25 scoring alone. Without semantic dense embeddings or cross-encoder reranking, retrieval suffered from severe vocabulary mismatch and synonym blindness. Queries with subtle semantic nuances achieved less than 35% context recall. The prototyping team formally abandoned single-stage keyword search in favor of the two-stage hybrid RRF pipeline.

Section 12: Disaster Recovery Verification & Backup Retention Schedule

Disaster recovery (DR) simulations are executed bi-weekly against active-passive database replicas. Continuous WAL (Write-Ahead Logging) archiving ensures continuous point-in-time recovery to any millisecond within a 35-day retention window. Automated health checks ping the cluster control plane every 500 ms; if three consecutive probes fail, DNS traffic fails over automatically.